INCH-POUND

MIL-C-70678 (AR)
AMENDMENT 4
11 March 1994
SUPERSEDING
AMENDMENT 3
01 July 1991

MILITARY SPECIFICATION

CASE, PROJECTILE BURSTER, XM216 (M216)

This Amendment forms a part of Military Specification MIL-C-70678 (AR), dated 14 January 1988, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "(M216)" and make this change wherever else the nomenclature appears in the specification.

1.1, line 3: Change "XM722" to M722" and make this change wherever else the nomenclature appears in the specification.

PAGE 2

2.2, Delete "ASTM E18 ... Rockwell Hardenss and Rockwell Superficial Hardness of Metallic Materials"

PAGE 4

Add new paragraph 3.7 as follows:

"3.7 Hydrotest. Prior to air leak testing selected burster casings shall be able to withstand the hydrostatic test load specified on Dwg. 9390728 without cracking, leakage, rupture or permanent deformation when tested in accordance with 4.5.5.

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PAGE 6

TABLE I, add the following under "Examination for defects".

"Hydrotest ...10 ...1/ ...3.7 ...4.5.5".

Add note "1/ Casings from forgings only, prior to protective finish. Perform air leak test on same units."

PAGE 8

4.4.2.1, Critical 3, delete "Visual/Gage; 6.2" and substitute "Gage".

Critical 5, delete "Presence of reverse taper on press fit surface and substitute, "Diameter of press fit surface increases while traversing from front to rear (open to closed) end of component."

Critical 7, delete in its entirety and substitute the following:

"7 ...Hydrotest ... 4.4.3.5 ...3.7 ...4.5.5/Gage."

Add Special "a ...Air test ...stamp missing ...100% ...3.2 ..Visual."

* Major 102, delete in its entirety.

Major 106, delete in its entirety.

PAGE 9

4.4.2.1, "Drawing Number" block: delete "9390278" and substitute "9390728."

Major 122, delete in its entirety.

PAGE 10

* 4.4.2.1, Add new Major Defect 130 and 131 as follows:

"130 Largest outside diameter ...0.40% ...3.2 ...Gage

131 Distance from face to basic diameter or O.D. ...0.40% ...3.2 ...Gage"

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Add Minor:

"207 Surface finish inside tube section improper ...5 ...3.2 ...Visual 1/ /6.2

208 Depth to basic diameter at forward inside chamfer ...0.65% ...3.2 ...Gage

209 Depth to basic diameter in chamfer rear of fuze booster cavity ...0.65% ...3.2 ...Gage."

Add note "1. Randomly selected casing shall be sectioned if a profilometer is unavailable. Any unit which fails the requirement shall cause the lot to be rejected."

PAGE 11

4.4.3.2, Delete in its entirety and substitute:

"4.4.3.2 Tension test. Four (4) tensile test specimens shall be fabricated from randomly selected locations from the raw material used (from each heat of steel) for each month's production. The test shall be in accordance with 4.5.3. This test only applies to casings machined from bar stock."

4.4.3.4, Delete in its entirety.

Add: "4.4.3.5 Hydrostatic test. Four (4) randomly chosen burster casings from the same mill heat of steel shall be selected from each day's production and tested in accordance with 4.5.5. This test only applies to burster casings produced from forgings."

PAGE 12

4.5.3, Delete in its entirety and substitute:

"4.5.3 Tension test. Material supplied in accordance with 4.4.3.2 shall be fabricated into the largest obtainable round tensile specimens as prescribed by ASTM Method E-8. Failure of any specimen to achieve any of the mechanical property requirements specified on Drawing 9390728 shall be cause for rejection of the lot represented by that heat of steel. The specimens shall be taken from the center of the bar. See 6.7."

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4.5.4, Delete in its entirety.

PAGE 12

Add:

"4.5.5 Hydrostatic test. After final machining but prior to the air leak test in 4.5.2, burster casings (manufactured from forgings only) supplied in accordance with Table I and 4.4.3.5 shall be tested using equipment capable of performing the test accurately and consistently. During the test, the pressure will be applied gradually until the required minimum pressure is reached and hold for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over four (4) hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of three (3) hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required by the test, the casings tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Casing rupture, cracking, leakage or bulging during the test (attributable to poor mechanical properties of the raw material) shall be cause for rejection of all material from the applicable heat of steel. Failure attributable to the production process shall be cause for rejection of all casings produced from expanding longitudinally or radially under pressure. Protective barriers should be installed to shield operators from metal fragments which may come from defective casings that fail the test. Examination should be made of all hydrotest plugs after a casing is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items. After successfully passing the hydrotest but prior to the air leak test, excess water shall be removed from the burster casings and each casing shall be dried at 250° F, minimum for 20 minutes, minimum. Casings which have been hydrotested shall not be returned to the production lot."

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Add:

"6.7 Tension test specimen preparation. It is recommended that a final finish of 60 to 125 be machined on the outside diameter of the specimens for accurate test results."

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